

Advanced techniques in applied economics

Lecture 6: Unobserved confounding and adjustments

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When the clean DAG picture breaks

What changes today?

In the previous lectures on DAGs we mostly acted as if the important variables were observed. Today we ask what happens when a relevant common cause is hidden.

Main difficulty

If an important confounder is unobserved, then

- back-door adjustment may fail,
- graphs on observed variables may be misleading,
- standard regression can be badly biased.

Main response. We cannot make hidden confounding disappear, but we can often partially adjust for it using

- instrumental variables,
- proxy controls,
- latent factor adjustments,
- model-based synthetic variation.

Main question: How can we reason causally when the graph contains hidden variables?

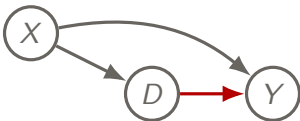
Roadmap for today

1. hidden confounding and why the clean DAG story breaks
2. a brief glimpse of MAGs
3. IV as the benchmark workaround
4. proxy controls
5. latent factor adjustments

Part 1: Hidden confounding

Recall: The clean adjustment story

Suppose we want the effect of treatment D on outcome Y .



Adjustment logic If X blocks all back-door paths from D to Y , then adjusting for X may identify the causal effect.

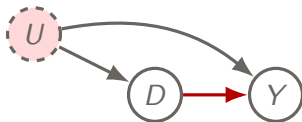
Ideal case

The confounder is observed, and the graph tells us what to condition on.

For the back-door criterion, see Pearl, *Causality*, Chapter 3.

What if the confounder is hidden?

Now suppose the true common cause U is unobserved.



Main problem

The back-door path

$$D \leftarrow U \rightarrow Y$$

is open, but we cannot condition on U because it is hidden.

Economic reading

U could be ability, firm quality, local demand, risk preferences, health, or political pressure.

A regression view of omitted-variable bias

Suppose

$$Y = \alpha D + h(W) + \lambda U + \varepsilon, \quad D = s(W) + \rho U + \nu,$$

where U is unobserved.

After partialling out W , the residualized equations are

$$\tilde{Y} = \alpha \tilde{D} + \lambda \tilde{U} + \varepsilon, \quad \tilde{D} = \rho \tilde{U} + \nu.$$

What goes wrong?

The remaining variation in D is correlated with the omitted term:

$$\text{Cov}(\tilde{D}, \lambda \tilde{U} + \varepsilon) = \lambda \rho \text{Var}(\tilde{U}) \neq 0.$$

Even after controlling for observed covariates, regression can remain biased.

For omitted-variable bias in regression language, see Wooldridge, *Econometric Analysis of Cross Section and Panel Data*.

A tiny omitted-variable simulation

Suppose

$$D = U + \varepsilon_D, \quad Y = \mathbf{1.0} D + U + \varepsilon_Y.$$

```
set.seed(1)
n <- 3000
U <- rnorm(n)
D <- U + rnorm(n)
Y <- 1.0 * D + U + rnorm(n)

coef(lm(Y ~ D))
```

```
# impossible benchmark:
# if U were observed
coef(lm(Y ~ D + U))
```

Interpretation

The naive regression on D alone mixes the true causal effect with hidden confounding from U .

Part 2: A brief glimpse of MAGs

Why introduce MAGs?

With latent variables, learning a DAG on the observed variables alone is not the right target.

What a MAG does

A **maximal ancestral graph** (MAG) is a mixed graph on the observed variables summarizing

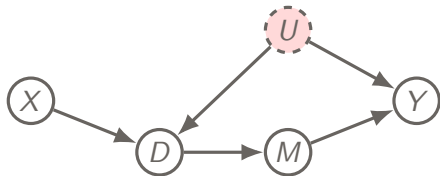
- directed relations that remain visible,
- dependence induced by hidden common causes,
- which conditional independences still hold among observed variables.

After marginalizing over hidden variables, the right graphical object is often not a DAG on the observed variables, but a mixed graph.

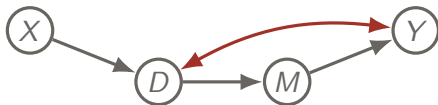
For MAGs and latent-variable graphical structure, see Richardson and Spirtes (2002), *Annals of Statistics*. For discovery with latent variables, see Zhang (2008) and Colombo et al. (2012), both in *Annals of Statistics*.

A four-variable example

Suppose the hidden DAG is



The corresponding observed mixed graph is



How to read it

- $X \rightarrow D \rightarrow M \rightarrow Y$ is the visible directed part,
- $D \leftrightarrow Y$ records hidden confounding between D and Y .

A conditional independence

One conditional independence encoded by this graph is $X \perp\!\!\!\perp M \mid D$. So MAGs still encode testable conditional independence restrictions, but now in a mixed graph.

What do we actually learn from data?

In practice, latent-variable discovery usually returns not a single MAG, but an equivalence class represented by a **PAG**.

Typical workflow

1. start from observed data only,
2. test conditional independences,
3. run FCI (Fast Causal Inference) or RFCI (Really Fast CI),
4. interpret the output as a PAG,

Why the output is harder to read

A PAG must represent several kinds of uncertainty:

- some directions are not identifiable,
- some dependence may come from hidden confounding,
- different MAGs may share the observed CI structure.

So a PAG is usually less decisive than a CPDAG, but also more honest about what the data do and do not identify.

The `pcalg` package implements `fci()` and `rfci()` for latent-variable discovery and works directly with PAG/MAG graph objects.

Why a DAG-based learner can be misleading

Consider the three-variable MAG



What the MAG encodes: $X \perp\!\!\!\perp Y$.

What PC does: In the population limit, PC recovers the same skeleton: $X - D - Y$.

Since X and Y are separated by $\emptyset$ and $D \notin \emptyset$, PC orients $X \rightarrow D \leftarrow Y$.

What went wrong?

The true graph says that D and Y share hidden confounding:

$$D \leftrightarrow Y.$$

A DAG learner cannot represent this, so it replaces latent confounding by an ordinary arrow.

So even with infinite data, PC can give a misleading causal picture when the true observed structure is Markov to a MAG rather than to a DAG.

A minimal R workflow with pcalg

```
library(pcalg)

suffStat <- list(C = cor(X), n = nrow(X))

fit <- rfci(suffStat,
            indepTest = gaussCItest,
            p = ncol(X),
            alpha = 0.01)
```

```
fit
plot(fit)

# for a smaller graph one can inspect
# the PAG adjacency matrix as well
amat(fit)
```

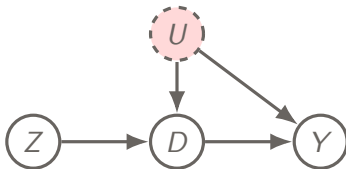
Interpretation

This is the latent-variable analogue of PC-style discovery: for Gaussian data we test conditional independences via partial correlations, but the output is a PAG rather than a DAG or CPDAG.

Part 3: IV as the benchmark workaround

IV intuition

If the confounder is hidden, the classical econometric workaround is to find an observed variable Z that moves the treatment but is otherwise exogenous.



Z is a valid instrument if:

- **Relevance:** Z affects D .
- **Exogeneity:** Z is independent of the hidden confounder.
- **Exclusion:** Z affects Y only through D .

Applied economics examples of IVs

Military service

Use draft lottery numbers to generate exogenous variation in service.

Education and earnings

Use distance to college, local college openings, or schooling reforms to shift education without directly shifting earnings.

Health and detention

Use physician or judge assignment when assignment is as-good-as random and decision-makers differ in strictness.

Main lesson

IV is often the cleanest benchmark when hidden confounding is serious and a credible instrument is available.

For an econometric treatment of IV, see Angrist and Pischke, *Mostly Harmless Econometrics*, or Wooldridge, *Introductory Econometrics*.

Why two-stage least squares works

Suppose the structural equation is $Y = \alpha D + \varepsilon$, where D is endogenous, so $\text{Cov}(D, \varepsilon) \neq 0$.

Instrument assumptions A valid instrument Z satisfies

$$\text{Cov}(Z, D) \neq 0, \quad \text{Cov}(Z, \varepsilon) = 0.$$

So Z is related to the treatment, but unrelated to the structural error.

Stage 1: isolate exogenous variation in D

Project D onto Z :

$$D = \pi Z + \text{residual}.$$

This gives fitted values $\hat{D}$, the part of D explained by the instrument.

Stage 2: estimate the effect

Regress Y on $\hat{D}$:

$$Y = \alpha \hat{D} + \text{error}.$$

This setting allows to incorporate other controls in the analysis.

Why this targets α

The variation in $\hat{D}$ is uncorrelated with ε , unlike the raw treatment D .

A tiny IV simulation

Suppose

$$D = 0.8Z + U + \varepsilon_D, \quad Y = \mathbf{1.0} D + U + \varepsilon_Y,$$

where Z is independent of U .

```
set.seed(1)
n <- 4000
Z <- rnorm(n)
U <- rnorm(n)
D <- 0.8*Z + U + rnorm(n)
Y <- 1.0*D + U + rnorm(n)
```

```
coef(lm(Y ~ D))
```

```
library(AER)
coef(ivreg(Y ~ D | Z))
```

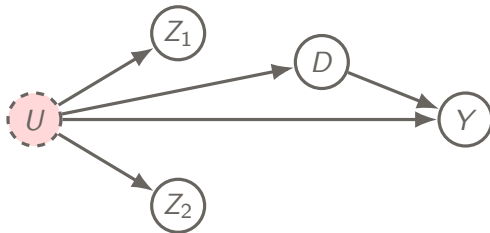
OLS is biased by the hidden confounder U . The command `ivreg(Y ~ D | Z)` implements 2SLS: it first extracts the part of D predicted by Z , then uses that variation to estimate the causal effect.

With heterogeneous treatment effects, IV generally identifies a local effect tied to the units whose treatment is moved by the instrument.

Part 4: Proxy controls and latent factor adjustment

The basic proxy idea

Even if we do not observe the confounder U , we may observe variables that are informative about it.



Idea

The proxies Z_1, Z_2 are not themselves the confounder, but they carry information about it.

Applied examples

Test scores, prior grades, neighborhood variables, baseline outcomes, or balance-sheet variables may all serve as noisy measurements of an unobserved confounder.

One proxy vs several proxies

A single noisy proxy is often not enough to remove confounding.

One proxy.

If $Z = U + \text{noise}$,
then conditioning on Z may reduce
confounding, but usually does not eliminate it.

Several proxies.

With multiple noisy measurements of the
same latent source, we can often extract a
better approximation to the hidden
confounder.

Proxy adjustment is usually an approximation strategy: it often reduces bias, but it does not automatically identify the causal effect.

For modern proxy-based identification ideas, see Miao, Geng, and Tchetgen Tchetgen (2018), *Biometrika*.

A tiny proxy-control simulation

Suppose

$$Z_1 = U + \eta_1, \quad Z_2 = U + \eta_2, \quad D = U + \varepsilon_D, \quad Y = 1.0 D + U + \varepsilon_Y.$$

```
set.seed(1)
n <- 3000
U <- rnorm(n)
Z1 <- U + rnorm(n)
Z2 <- U + rnorm(n)
D <- U + rnorm(n)
Y <- 1.0*D + U + rnorm(n)

coef(lm(Y ~ D))
```

```
coef(lm(Y ~ D + Z1))
coef(lm(Y ~ D + Z1 + Z2))
```

Interpretation

One proxy helps a bit; multiple proxies help more. The point is not exact identification, but partial recovery of the hidden confounder and partial bias reduction.

Latent factor adjustments

Structured proxy adjustment

Suppose many observed variables are all affected by the same hidden source U . Instead of using them one by one as proxy controls, we summarize them through a low-dimensional factor model.

Adjustment strategy

Estimate one or a few latent factors, then include those estimated factors as additional controls in the outcome model.

Economic reading

This is natural when hidden confounding comes from business-cycle conditions, latent productivity, local demand, or market-wide financial stress.

For interactive effects and latent-factor adjustments in panel settings, see Bai (2009), *Econometrica*.

A tiny latent-factor adjustment in R

```
set.seed(1)
n <- 2500
U <- rnorm(n)

X1 <- 0.8*U + rnorm(n)
X2 <- 0.7*U + rnorm(n)
X3 <- 0.9*U + rnorm(n)
D <- U + rnorm(n)
Y <- 1.0*D + U + rnorm(n)

X <- cbind(X1, X2, X3)
```

```
pc1 <- prcomp(X, scale. = TRUE)$x[,1]

coef(lm(Y ~ D))
coef(lm(Y ~ D + pc1))
```

Interpretation

The first principal component acts as a rough estimate of the hidden confounder and can reduce omitted-variable bias.

Take-away

- Hidden confounders break the clean adjustment logic of observed-variable DAGs.
- MAGs give a compact language for the observed implications of latent confounding.
- IV is the benchmark workaround when a credible instrument is available.
- Proxy controls use noisy measurements of the hidden confounder.
- Latent factor adjustments use many observed variables to estimate a hidden source.

When the graph is only partially observed, the goal shifts from perfect recovery to careful partial adjustment.

Looking ahead

Next lecture

We move from hidden confounding and partial adjustment to a different kind of structure: positive dependence and total positivity.

Bridge forward

Today the main difficulty was that hidden variables distort causal interpretation. Next time we will see that even without a causal story, the *sign* of dependence can itself be useful structure for modeling and estimation.